

# Aleksandr Krasnok, Ph.D.

Assistant Professor, Electrical & Computer Engineering (FIU) | Director, Quantum Technologies & Metamaterials (QTM) Lab

IEEE Senior Member | Courtesy Appointment, Knight Foundation School of Computing and Information Sciences

Quantum Control • Superconducting Circuits • Cryogenic Systems • Quantum Sensing

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## Academic Profile

Assistant Professor of Electrical and Computer Engineering and Director of the QTM Lab at Florida International University. Research in quantum-hardware control, circuit and cavity QED, cryogenic photonics, and quantum sensing. Co-founder of the FIU Quantum Initiative and core member of FIU's university-wide quantum committee. Developed and taught the undergraduate and graduate quantum-computing sequence.

**Career-wide research record:** 200+ scholarly works; **15,222 citations**, ***h*-index 61**, ***i*<sub>10</sub>-index 113** (Google Scholar, September 25, 2026).

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## Professional Preparation and Appointments

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**Ph.D., Photonics and Quantum Optics**, ITMO University, 2013.

**M.S., Quantum Optics**, Far Eastern Federal University, 2010.

**B.S., Quantum Optics**, Far Eastern Federal University, 2008.

- **Assistant Professor, Electrical & Computer Engineering, FIU**, August 2021–present. Director, QTM Lab; courtesy appointment, KFSCIS. **45+ journal publications since joining FIU**, including *Science* and *Physical Review Letters*. Collaborate on quantum circuits, superconducting resonators, and plasma experiments with FIU/CUNY, RadiaBeam, Chicago and Argonne (USA), Calgary/Simon Fraser (Canada), and LAPLACE–Toulouse (France). Co-founded FIU’s quantum initiative and developed its quantum-computing course sequence.
- **Research Assistant Professor; Founding Director, Photonics Core Facility, CUNY ASRC**, July 2018–July 2021. Established shared-user operations from the ground up: approximately **eight optical tables, 100+ users by 2021**, and a **\$162,200 FY2019 facility budget**. Capabilities included femtosecond lasers, harmonic generation, supercontinuum spectroscopy, and infrared ellipsometry. **30+ journal publications during this appointment**; *Nature* photonic twistrionics, a *Physics World* Top 10 Breakthrough of 2020, with NUS (Singapore) and Monash (Australia).
- **Postdoctoral Researcher / Research Scientist, UT Austin**, November 2016–July 2018. **20+ journal publications during this appointment** in Andrea Alù’s group. Connected theory with experiments in quantum-dot control, exciton–plasmon coupling, and coherent power transfer through collaborations with Zheng/UT Austin, Wu/South Carolina and Bracker/NRL (USA), Aalto (Finland), and Chalmers (Sweden).
- **Senior Research Fellow, Photonics / Electrodynamics, ITMO**, June 2013–November 2016. Led an all-dielectric nanophotonics group of **two postdocs and approximately five students; 20+ journal publications during this appointment**. Developed quantum-emitter/nanoantenna and optical-resonance research with ITMO experimental teams, Kivshar/ANU (Australia), and Simovski–Tretyakov/Aalto (Finland).

**Institutional context.** FIU rose from No. 78 to No. 46 among U.S. public universities in *U.S. News & World Report* (2022 edition; 2027 edition).

## Quantum Research and Development

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- **Qubit control and hardware diagnostics.** Complex-frequency driving, scattering zeros, and shaped pulses for selective excitation and crosstalk suppression in coupled superconducting qubits and resonators.
- **Circuit and cavity QED.** Fano interference, dark states, quantum nonreciprocity, superconducting microwave cavities, and Purcell-decay protection.
- **Cryogenic quantum systems.** Dielectric resonators, antennas, packaging, and photonic interfaces for sensing and signal transfer across cryogenic temperature stages.
- **Quantum sensing.** NV-diamond magnetometry, resonant sensing, and quantum magnetometers for metrology and infrastructure monitoring.
- **Quantum photonics.** Light–matter coupling, non-Hermitian dynamics, embedded quantum states, and microwave–optical interfaces.

## Quantum Leadership and Partnerships

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- **Co-Founder, FIU Quantum Initiative**, 2025–present, with Prof. Janki Bhimani. Coordinate faculty research, curriculum, seminars, and outreach.
- **Core Member, Committee on FIU Initiatives in Quantum Science, Education, and Industry Partnerships**, 2026–present. Contribute expertise in quantum hardware, photonics, sensing, and engineering education.
- **FIU–IonQ initiative**, 2026–present. Participated in FIU’s faculty/leadership visit to IonQ and contribute to research and educational planning for the **IonQ Superion 256 trapped-ion quantum computer**. FIU announced the agreement in September 2026; campus arrival is expected in late 2027. [FIU announcement](#).
- **Industry and statewide engagement**. Faculty representative, FIU Quantum AI Showcase at eMerge Americas (2026); organizer of FIU participation in Quantum Beach (2025); participant in FIU–Microsoft quantum discussions (June 2026).

## Selected Research Funding and Computing Resources

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### Active Research Awards

- **DOE / Brookhaven National Laboratory: El-Pho**. FIU subcontract **PI**; Electro-Photonics Integrated Platform for Near-Sensor Processing in Extreme Environments; subcontract 463609. **\$600,000 ceiling; \$60,000 initially allotted**, with incremental funding. Base period March–September 2026; annual options through September 30, 2029.
- **AFOSR: Complex Frequency Excitations in Natural and Artificial Plasma**. **PI**; **\$50,000**; June 1–November 30, 2026; FA9550-26-1-B070.
- **NSF I-Corps: Meta-MRI**. **PI**; **\$50,000**; July 1, 2024–December 31, 2026; award 2433854. Translation of electromagnetic wave manipulation to magnetic resonance imaging.

### Institutional Start-up Support

- **FIU faculty start-up package**, 2021: **\$500,000** for research and infrastructure, plus separate tuition and stipend support for **three graduate students over two academic years**.

### Completed Support and Computing Resources

- **IC Postdoctoral Research Fellowship: Quantum Sensors for Optical Readout of Superconducting Qubits**. Research Advisor / Host for Dr. Ingrid Torres; October 2023–January 14, 2026.
- **DOE SBIR / RadiaBeam: Ferrite-Free Circulator for Precise Measurements of SRF Cavities with High Q-Factor**. Co-PI / FIU lead; **\$63,000 to FIU**; 2022–2023.
- **NVIDIA Academic Grant Program: CFQ-Control**. **PI**; **32,000 A100 80GB GPU-hours** via NVIDIA Brev; January–June 2026.
- **IonQ cloud quantum-computing access** for teaching and research.

## Selected Scientific Publications

### Book

1. **A. Krasnok**, *Introduction to Quantum Computing and Sensing for Electrical Engineers: A Hands-On Guide*. Synthesis Lectures on Emerging Engineering Technologies, Springer, Cham, first edition, forthcoming 2026. Hardcover ISBN: 978-3-032-34529-5; eBook ISBN: 978-3-032-34530-1. [Springer book page](#).

### Journal Articles

1. D. Trivedi, A. Bhowmik, L. Niaz, and **A. Krasnok**, "Reflection-Zero Waveforms for Selective Loading in Shared-Line Superconducting Qubit Circuits," *Advanced Quantum Technologies* **9**(9), e70441 (2026). doi:10.1002/qute.70441. Earlier version: arXiv:2506.03316. *Selected for a cover feature*.
2. M. M. Hasan, I. Torres, and **A. Krasnok**, "Quantum Magnetometers for Infrastructure Inspection and Monitoring," *Measurement* **290**, 122735 (2026), doi:10.1016/j.measurement.2026.122735.
3. **A. Krasnok**, "What Quantum Computer to Buy?" *AVS Quantum Science* **8**(3), 034101 (2026). doi:10.1116/5.0337237. Preprint: arXiv:2604.04761.
4. **A. Krasnok** and D. Seletskiy, "Complex-Frequency Chirped Pulses for Trajectory-Resolved Scattering," *ACS Photonics* (2026), doi:10.1021/acsp Photonics.6c00602.
5. **A. Krasnok**, "Constant-Amplitude  $2\pi$  Phase Modulation from Topological Pole–Zero Winding," *Physical Review Letters* **137**(5), 053801 (2026), doi:10.1103/xt46-rfjs.
6. **A. Krasnok**, A. Sarwat, W. Anderson Jr., and A. Gil, "Universities Need Transparent Quantum Hardware," *Nature Reviews Physics* (2026) [Comment], published July 15, 2026, doi:10.1038/s42254-026-00963-4.
7. **A. Krasnok**, "Metamaterials in Superconducting and Cryogenic Quantum Technologies," *Applied Physics Reviews* **13**, 021311 (2026), Special Collection: *Materials for Superconducting Quantum Devices*, doi:10.1063/5.0282013.
8. **A. Krasnok** and X. Xu, "Editorial: Emerging Trends in Photonic Quantum Computing," *Advanced Photonics* **8**(1), 010102 (2026), doi:10.1117/1.AP8.1.010102.
9. I. Torres and **A. Krasnok**, "A cryogenic dielectric antenna for wireless sensing and interfacing outside the 10 K environment," *Cryogenics* **155**, 104286 (2026), doi:10.1016/j.cryogenics.2026.104286.
10. D. Trivedi, L. Niaz, A. Alù, and **A. Krasnok**, "Selective Excitation of Coupled Resonators via Complex Frequency Driving: Enhanced Efficiency and Crosstalk Suppression," *ACS Photonics* **12** (11), 5929–5938 (2025), doi:10.1021/acsp Photonics.5c01244.
11. D. Trivedi, A. Madanayake, and **A. Krasnok**, "Revealing Invisible Scattering Poles with Complex-Frequency Signals," *Journal of Applied Physics* **137** (24), 243103 (2025), doi:10.1063/5.0262971.
12. S. Kim, **A. Krasnok**, and A. Alù, "Complex-Frequency Excitations in Photonics and Wave Physics," *Science* **387**(6741), eado4128 (2025), doi:10.1126/science.ado4128.
13. C. Zhang, M. Kim, Y.-H. Zhang, Y.-P. Wang, D. Trivedi, **A. Krasnok**, J. Wang, D. Isleifson, R. Roshko, and C.-M. Hu, "Gain–Loss Coupled Systems," *APL Quantum* **2** (1), 011501 (2025), doi:10.1063/5.0250178. *Front cover*.



14. L. Niaz and **A. Krasnok**, “Quantum Leap: Observing Antiferromagnetic Transition in a 3D Fermionic Hubbard Model with Ultracold Atoms,” *Advanced Photonics* **6**(6), 060503 (2024), doi:10.1117/1.AP.6.6.060503.
15. D. Trivedi, A. Madanayake, and **A. Krasnok**, “Anomalies in Light Scattering: A Circuit-Model Approach,” *Physical Review Applied* **22**(3), 034061 (2024), doi:10.1103/PhysRevApplied.22.034061.
16. **A. Krasnok**, P. Dhakal, A. Fedorov, P. Frigola, M. Kelly, and S. Kutsaev, “Superconducting Microwave Cavities and Qubits for Quantum Information Systems,” *Applied Physics Reviews* **11**(1), 011302 (2024), doi:10.1063/5.0155213.
17. S. V. Kutsaev, R. Agustsson, A. C. Araujo-Martinez, D. M. Broun, P. Carriere, M. H. Chou, T. S. Chouinard, A. N. Cleland, P. Frigola, M. P. Kelly, **A. Krasnok**, N. Matavalam, A. Moro, R. G. Povey, T. Reid, and A. Y. Smirnov, “Enhancing the Q-Factor of a Practical Qubit Niobium Three-Dimensional  $\lambda/4$ -Resonator Through Surface Treatment,” *IEEE Transactions on Applied Superconductivity* **34**(2), 1700909 (2024), doi:10.1109/TASC.2024.3354948.
18. D. Trivedi, L. Belostotski, A. Madanayake, and **A. Krasnok**, “Fano-Qubits for Quantum Devices with Enhanced Isolation and Bandwidth,” *Applied Physics Letters* **122**, 264003 (2023), doi:10.1063/5.0151047.
19. N. Nefedkin, M. Cotrufo, **A. Krasnok**, and A. Alù, “Dark-State Induced Quantum Nonreciprocity,” *Advanced Quantum Technologies* **5**(3), 2100112 (2022), doi:10.1002/qute.202100112.
20. N. Nefedkin, A. Alù, and **A. Krasnok**, “Quantum Embedded Superstates,” *Advanced Quantum Technologies* **4**(6), 2000121 (2021), doi:10.1002/qute.202000121.
21. S. V. Kutsaev, **A. Krasnok**, S. N. Romanenko, A. Yu. Smirnov, K. Taletski, and V. P. Yakovlev, “Up-and-Coming Advances in Optical and Microwave Nonreciprocity: From Classical to Quantum Realm,” *Advanced Photonics Research* **2**(3), 2000104 (2021), doi:10.1002/adpr.202000104.
22. A. E. Krasnok *et al.*, “Enhanced Emission Extraction and Selective Excitation of NV Centers with All-Dielectric Nanoantennas,” *Laser & Photonics Reviews* **9**(4), 385–391 (2015). doi:10.1002/lpor.201400453.

## Preprints and Submitted Manuscripts

1. **A. Krasnok**, “Which pulse maximizes resonant nonlinear conversion?” arXiv:2608.19464 (2026). Submitted to *Physical Review Applied*, September 2026.
2. **A. Krasnok**, “What is a quantum complex scattering zero?” Submitted to *Quantum Science and Technology*, August 16, 2026.
3. **A. Krasnok**, “Quantum Computing Courses Need Real Machines.” *Communications of the ACM*, under consideration.
4. **A. Krasnok**, “Protecting Qubits from Purcell Decay via Permanent Dipoles,” *arXiv preprint* arXiv:2606.22732 (2026), doi:10.48550/arXiv.2606.22732.

## Conference Proceedings

1. I. Torres and **A. Krasnok**, “Cryogenic Dielectric Mie Resonators for Quantum Applications,” *Proc. SPIE* **13379**, *Photonic Heat Engines: Science and Applications VII*, 133790H (2025), doi:10.1117/12.3052180.
2. D. Trivedi, A. Madanayake, L. Belostotski, and **A. Krasnok**, “Enhancing Quantum Non-Reciprocity: From Lorentz to Fano Qubits for Improved Isolation and Bandwidth in Circuit QED Systems,” *Proc. Applied Computational Electromagnetics Society (ACES) Symposium*, Orlando, FL, USA, paper 45-04 (2024).

## Science Communication

- **A. Krasnok**, “Universities should judge quantum-computing investments by what students learn,” *Physics World*, September 9, 2026. [Article](#).
- **A. Krasnok**, “Quantum sensors could spot hidden damage in the thousands of US bridges rated structurally deficient,” *The Conversation*, June 19, 2026. Republished by *Scientific American* and *FIU News*. [Article](#).
- **A. Krasnok**, “Cryogenic photonics gives superconducting quantum hardware a path to scale,” *Laser Focus World*, May 7, 2026. [Article](#).

## Intellectual Property

- **A. Krasnok**, D. Trivedi, and L. Niaz, “Systems and Methods for Selective Control of Coupled Quantum Elements.” U.S. Patent Application Serial No. 19/694,219, filed June 1, 2026; FIU disclosure 2025-043; Wellspring record D2025-0055.
- **A. Krasnok**, “Dressed-Rate-Optimized Permanent-Dipole Purcell Protection with Longitudinal Qubit Readout.” FIU invention disclosure D2026-0073 (2026; under review).
- **A. Krasnok**, “Trajectory-Resolved Complex-Frequency Scattering Measurement with Chirped Analytic Pulses.” FIU invention disclosure D2026-0075 (2026; under review).

## Quantum Education and Mentorship

### Teaching and Open Resources

- Developed and taught FIU’s **EEE 4423 Introduction to Quantum Computers** and **EEE 6429 Advanced Quantum Computers**, using Qiskit, Python, quantum circuits, and hardware exercises. Both courses received the *Affordability Counts* medallion for eliminating required textbook costs.
- Taught graduate special topics in **Quantum Technologies and Materials** and **Quantum Nanophotonics**, including quantum sensing, communication, superconductors, and non-Hermitian scattering.
- Published **70+ recorded modules** and course playlists covering quantum computing, sensing, and computational electromagnetics. [QTM Lab YouTube channel](#).
- Administered the **Quantum Computing Conceptual Survey** in EEE 4423 in Spring 2026, with comparisons to 878 students across 59 courses at 46 institutions.

### Mentorship

Supervised three postdoctoral fellows, co-supervised one Ph.D. to completion, and mentored 11 undergraduate researchers and Senior Design students at FIU. Build a respectful, supportive research environment through open feedback, technical training, and career guidance, including support for moves to new academic and industry positions.

- **Current graduate researchers:** **Aditya Bhowmik** (2025–present), **Md. Alim Uddin** (May 2026–present), and **Md. Nakibul Islam** (August 2026–present). Research in quantum photonics, complex-frequency wave physics, and sensing.
- **Laraib Niaz**, research collaborator and Ph.D. research mentee, 2024–present. FIU Ph.D. readmission pending; continuing remote work on quantum simulations and complex-frequency control.

- **Deepanshu Trivedi**, Ph.D., FIU, December 2025; co-supervised with Prof. Arjuna Madanayake. Research on complex-frequency control and Fano-qubits. Now Design Engineer, **imec USA Florida, Superconducting Electronics team**.
- **Graduate alumnus: Md. Ibrahim Khalil**, FIU, 2025–January 2026. Research on silicon microring resonators and free-carrier effects; transferred to another university in Spring 2026.
- **Ingrid Torres**, Postdoctoral Fellow, FIU. Research on quantum sensing, cryogenic dielectric resonators, nanofabrication, and cryogenic wireless sensing.
- **Muhammad Mahmudul Hasan**, Postdoctoral Research Associate, FIU, 2025–2026. Research on quantum sensors, diamond TeraFETs, and THz photonics; joined **Nokia** in August 2026.
- **Undergraduate researchers: Nafiul Khalid**, FIU Honors College, 2025–present: Qiskit/PennyLane, quantum machine learning, and quantum-computing learning resources. **Quesnay Lundy**, 2023–2024: wireless power transfer and CST simulations; continued studies at the University of Florida.
- **Senior Design, 2026–present: Handheld Moisture Tomography Device for Wall Inspection**. Students: **Sean Avendano, Andy Lara, Pablo Lucero, and Degle Martinez**. Faculty mentor for radar-based moisture mapping behind drywall; Senior Design I, Spring 2026, and II, Fall 2026.
- **Senior Design, Fall 2023–Spring 2024: Low-Cost Air Quality Kit**. Students: **Caio Moreti Koga, Jasim Bhatti, Adrian Ortega, Andres Lara, and Nurain Chowdhury**. Faculty mentor for development and testing of a low-cost air-quality monitoring system; completed.

## Selected Talks and Seminars

- **September 24, 2026**, Missouri State University, PAMS Seminar (virtual): “Quantum Computing through the Lens of Physics.”
- **February 20, 2026**, imec USA Florida, Kissimmee, FL: “Cryogenic Quantum Systems Engineering: From mK Qubits to Practical RF/Photonics Control.”
- **February 19, 2026**, FIU Dean’s Leadership Council: “Quantum-Enabled Resilience: Infrastructure Health Sensing and Quantum-Assisted Optimization from Models to Prototypes and Testing.”
- **December 2, 2025**, University of North Texas: “Engineering Complex-Frequency Excitations: From Virtual Gain in Photonics to Selective Qubit Control.”
- **November 20, 2025**, FIU CEC University–Industry Collaboration Session: “Quantum Photonics for Extreme Environments and Real-World Sensing.”
- **August 18, 2025**, Florida Alliance for Quantum Technology / Florida Quantum Initiative Meeting, University of South Florida: “Quantum Sensing for Infrastructure Resilience: Testing at FIU’s Wall of Wind.”
- **April 17, 2025**, Florida Quantum Summit, University of Florida: “From Quantum Light to Student Might: How FIU is Teaching and Building the Future of Sensing Technology.”
- **December 12, 2022**, Polytechnique Montréal: “Quantum Nonreciprocity with Nonlinearity and Weyl Semimetals.”

## Editorial Service and Community Organization

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- **Associate Editor**, *Photonics Research*, 2022–present. **Editorial Board Member**, *Advanced Photonics Nexus*, 2024–present, and *Reviews on Advanced Materials Science*, 2023–present.
- **Guest Editor**, “Emerging Trends in Photonic Quantum Computing,” *Advanced Photonics* and *Advanced Photonics Nexus*, 2025–2026, with Prof. Xiulai Xu.
- **Organizer and Host**, FIU Qiskit Fall Fest, November 8–10, 2025; workshops, challenges, mentoring, and invited speakers.
- **Organizer and Session Co-Chair**, “Complex Frequency Excitation Effects in Photonics and Wave Physics,” IEEE RAPID 2025, with Seunghwi Kim.
- **Session Co-Chair**, “Quantum Techniques for Electromagnetics,” URSI AT-AP-RASC, June 2, 2022. **Panelist and Mentor**, IBM Quantum Computing Fall Fest, Miami Dade College, November 2024.
- Journal reviewer in quantum technologies and photonics; proposal reviewer for DOE and NSF.

## Selected Honors and Recognition

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- **Rising Star in Engineering**, Academy of Science, Engineering and Medicine of Florida (ASEMFL), 2026.
- **Selected, Fulbright Specialist Roster**, U.S. Department of State; August 14, 2026–August 14, 2029.
- **Finalist, FIU Presidential Excellence Award**, 2026.
- **FIU Top Scholar Award, Research**, 2024–2025.
- **Leopold B. Felsen Award for Excellence in Electrodynamics**, EurAAP, 2024.
- **APL Rising Stars**, *Applied Physics Letters*, 2023; “Fano-qubits for quantum devices with enhanced isolation and bandwidth.”

## Technical Skills

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- **Quantum modeling and control**: Lindblad dynamics, circuit/cavity QED, pulse shaping, scattering poles and zeros; QuTiP, Qiskit, Python, and GPU/HPC computing.
- **Electromagnetic simulation**: FEM, FDTD, MoM, resonator and circuit models; COMSOL, Ansys HFSS, CST, Keysight ADS, and MATLAB.
- **Experimental methods**: microwave resonator measurements, VNA/S-parameters, cryogenic and RF workflows, free-space and fiber optics, PL/Raman spectroscopy, and Python instrument control.